The Langlands conjectures predict that the Frobenius classes attached to an automorphic representation behave like Haar random elements of a compact group. This book studies that probabilistic shadow: what it proves, what it tests, and where it stops.

Part by part it develops the Weyl criterion, the Sato Tate law and murmurations of root numbers; Hecke operators as random walks on Bruhat Tits trees, Ihara zeta functions, Ramanujan graphs and hitting times on supersingular isogeny graphs; infinitely divisible laws, generalized gamma convolutions and Thorin measures for L functions, and in the same language squarefree integers, Dirichlet's beta function, quadratic reciprocity and Fermat's Last Theorem; inverse Laplace transforms that pass between the faces of one L function; random integers and the Poisson Dirichlet law; Dirichlet and Hecke characters and complex multiplication; Maass forms and Brownian motion on the modular surface; crossing probabilities in critical percolation and the L function of an elliptic curve; even Galois representations, torsion and potential modularity; an extreme value representation of the L functions of eta products; and Bayesian identification of Sato Tate groups.

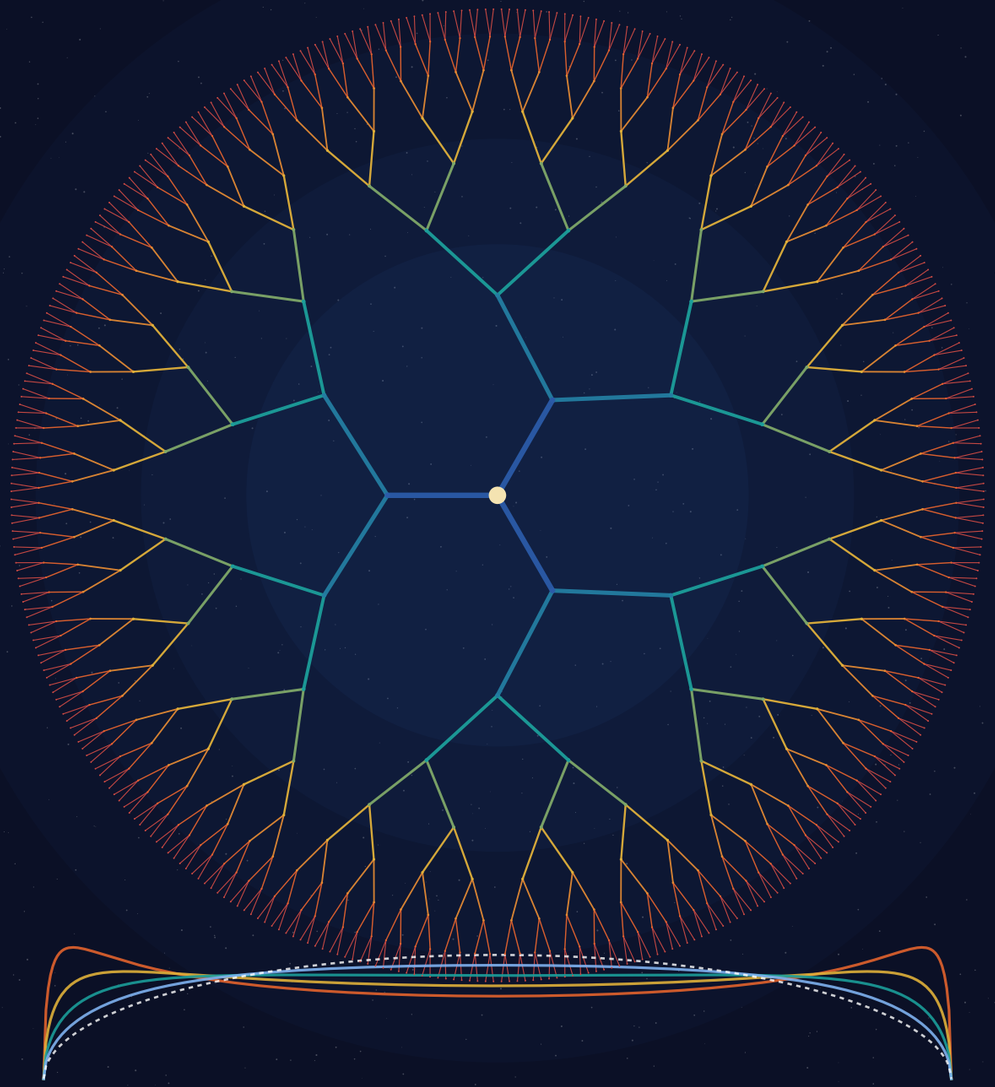
New results include an exact limit for the Kemeny constants of isogeny graphs, an equivalence between the Riemann hypothesis for Ramanujan's Delta and a generalized gamma convolution, and the identity L(s,f) = E[Q(s, 2πM)] expressing L functions as expectations. Large scale computations accompany every chapter, and each chapter ends with open computational problems that together form a research programme.

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Probabilistic Methods



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